

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

November 2013

EC 407 Radar Systems

Date: 23.11.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Enlist types of navigational aids. [01]
- (b) With a help of antenna patterns explain the working of sense finder in Direction Finding. [06]
- Q - 2 (a) What is receiver noise? Derive the Radar range equation in terms of bandwidth. [05]
- (b) Calculate power of minimum detectable signal in a radar receiver operating in standard ambient temperature of 17°C , which has an IF bandwidth of 3 MHz and a Noise figure of 6 dB. [04]
- (c) What are the important applications of Radar? Also described Frequency band used for various applications. [05]

OR

- Q - 2 (a) What types of displays will be used in Radar? Explain any one in detail. [05]
- (b) Derive Basic Radar Equation. [04]
- (c) If Radar is designed for operation at 10GHz with an antenna of diameter 2 m and minimum detectable signal of $36 \times 10^{-15} \text{W}$, calculate the peak power required to have maximum range of 1000 Km with a target of cross sectional area 20m^2 . Use $A = \frac{\pi D^2}{4}$ [05]
- Q - 3 (a) Explain filter characteristics of Delay line canceller. [05]
- (b) What do you understand by blind speed? How to eliminate it. [05]
- (c) Two MTI radars are found to have a blind speed are given by $V_{b2}=20 \text{m/s}$ of first MTI radar and second Radar at $V_{b3}=30 \text{ m/s}$ of second MTI radar if they operate on wavelength of 30 cm find the ratio of PRF of the Radar. [04]

OR

- Q - 3 (a) Explain how multiple staggered PRF can be used to process with a transversal filter. [05]
- (b) Explain Coherent MTI radar with a help of a block diagram. What is the need of COHO and STALO? [05]
- (c) An MTI radar is operating with 10Ghz with a PRF of 1000 Hz. Calculate the lower three blind speeds. [04]

SECTION – II

- Q - 4 (a) Explain Principle of Hyperbolic Navigation. [02]
(b) How Loop antenna can be used for direction finding? What is the problem of direction ambiguity? How can we solve this problem? [05]
- Q - 5 (a) Explain AIR borne DME interrogator with modes of operation. Compare DME and TACAN. [07]
(b) Write a Short note on [07]
(a) Remote Sensing Radar (b) SAR Radar

OR

- Q - 5 (a) What is Doppler navigation? Explain Doppler effect and derive the equation [07]
$$f_D = \frac{2V_r}{\lambda}$$
 Explain any one type of beam configuration used in Doppler navigation.
(b) Write a Short note on [07]
(a) MET Radar (b) ASR Radar
- Q - 6 (a) Draw neat block diagram of FMCW radar and enlist its distinguished features. [07]
(b) Draw the block diagram of FMCW altimeter and explain it. [07]

OR

- Q - 6 (a) How detection of multiple targets moving with different velocities has done by Doppler Radar? [07]
(b) Explain the Pulse Doppler radar with neat block diagram. List the advantages of it over CW radar. [07]